

Spring26-Assignment1 Simple Query

Question 1:

In the stations table, find the english_name of stations containing "ROAD, HILL" (including uppercase and lowercase letters), such as Hanghai Road, Antuo Hill, etc., and output the content in the following format. The output results are sorted in ascending order by station_id.

在stations表格中找到英文名中带有 ROAD, HILL (包含大小写)等车站(如: Hanghai Road, Antuo Hill), 按如下格式输出内容。输出结果根据 station_id 升序排列。

```
select english_name, chinese_name
from stations
where upper(english_name) like '%ROAD%'
      or upper(english_name)like  '%HILL%'
order by station_id;
```

english_name	chinese_name
Ma'an Hill	马安山
Antuo Hill	安托山
Mission Hills	观澜湖
Hanghai Road	航海路
Zhenhai Road	振海路
Linhai Road	临海路
Qianhai Road	前海路
Xuefu Road	学府路

Question 2:

In the Station table, calculate the percentage of the total number of stations in Nanshan District, Futian District, and Luohu District is in the total number of all stations? Please give your answer with two decimal places and concatenate "%".

在Station表中, 南山区、福田区、罗湖区的地铁站数量占地铁站总数量的百分比? 答案请保留两位小数并拼接"%".

```
select Round(100.0 * sum(count) / count(*) ,2) || '%' percentage
from (select case district
              when 'Nanshan' then 1
              when 'Luohu' then 1
              when 'Futian' then 1
              else 0 end count
       from stations) sub
```

percentage
65.78%

Question 3:

Using the table **bus_lines**, which bus lines have more than 20 (included) stations, and how many stations they have, please output results in descending order

周围公交站数量大于20的地铁站有多少个？

```
select count(*) from (
select station_id, count(*) as busCnt from bus_lines
group by station_id
having count(*)> 20) sub;
```

Count
61

Question 4:

Among all subway stations with bus stops nearby, find the `station_id` of the subway station with the fewest surrounding bus stops, and output these `station_id` in ascending order. If multiple subway stations have the same number of surrounding bus stops, find them all as well.

在所有周围有公交站的地铁站中，找到周围公交站数量最少的地铁站的id，并升序输出这些地铁站的id。如果多个地铁站周围公交站数量并列最少，也都要找出来。

```

with q as (select station_id, count(*) as busCnt
            from bus_lines
            group by station_id)
select station_id
from q
where q.busCnt = (select min(busCnt) from q)
order by station_id;

```

Station id
23
26
54
120
179
181

Question 5

Find all transfer stations on Metro Line 5 and output their `english_name` in ascending order of their `station_id`.

找出地铁五号线上的所有中转站，按照 `station_id` 升序 输出这些中转地铁站的英文名

```

select s.english_name
from (select station_id, count(*)
      from line_detail ld
      where station_id in (select station_id
                           from line_detail ld
                           where ld.line_id = 5)
      group by station_id
      having count(*) > 1) sub
join stations s on s.station_id = sub.station_id
order by s.station_id;

```

english_name
Qianhaiwan
Bao'an Center
Huangbeiling
Buji
Shenzhen North Station
Xili
Tai'an

Question 6:

How many stations are there in Longhua District that are not on No. 5 line?

在龙华区的地铁站，但不在地铁五号线上的有多少个？

```
select count(*)
from (select s.station_id
      from stations s
      where s.district = 'Longhua') s1
left join(select station_id
          from line_detail ld
          where line_id = 5) s2 on s1.station_id = s2.station_id
where s2.station_id is null;
```

count
7

Question 7:

Find the terminus station of subway lines 1, 2, and 3, which is the subway station with the largest number (num). List the line_id and their corresponding english_name in ascending order of line_id.

找到1, 2, 3号地铁线的终点站，即num最大的地铁站。按照line_id的升序，列出line_id 与地铁站的英文名。

```

select ld.line_id, s.english_name
from line_detail ld
      join (select line_id, max(num) as max
            from line_detail ld
            where line_id in (1, 2, 3)
            group by line_id) sub on sub.max = ld.num and sub.line_id =
ld.line_id
join stations s on ld.station_id = s.station_id
order by line_id;

```

line_id	english_name
1	Airport East
2	Xinxiu
3	Shuanglong

Question 8:

Find the lines whose contains most stations, and then return the fourth (num=4) stations in those lines. If your result set contains more than one rows, output them all in **ascending** order by their Chinese names. Your output should contain the metro line id, station id and the Chinese name.

找到拥有最多地铁站的地铁线，并返回它的第四个地铁站。如果结果集中有多行数据，那么根据地铁站中文名升序排列。返回结果集中的列包含line id, station id与中文名。

```

select l.line_id, l.station_id, s.chinese_name
from (with q as (select count(*) cnt, line_id
                 from line_detail
                 group by line_id)
      select line_id, cnt
      from q
      where cnt =
            (select max(cnt)
             from q)) x
      join line_detail l on x.line_id = l.line_id
      join stations s on l.station_id = s.station_id
where l.num = 4
order by s.chinese_name;

```

line_id	station_id	chinese_name
9	165	前海路

